Projet OCaml: Optimal Reciprocal Collision Avoidance

Richard ALLIGIER Ecole Nationale de l'Aviation Civile
Toulouse, FRANCE
richard.alligier@enac.fr

I. Introduction en Français

Le but de ce projet est d'implémenter l'algorithme Optimal Reciprocal Collision Avoidance (ORCA). Cet algorithme issu de la robotique permet à des agents (les robots) d'éviter d'entrer en collision. Il ne nécessite pas de communication "explicite" entre agents. Les seules informations "communiquées" sont la position et la vitesse des autres agents.

On va implémenter cet algorithme dans un contexte de traffic aérien, en deux dimensions en supposant que tous les avions sont à la même altitude. Chaque avion devra être séparé des autres de 5NM. Pour garantir cette séparation ORCA va contraindre les vitesses relatives entre chaque paire d'avions. Dit triviallement avec des approximations: la vitesse relative ne doit pas pointer vers l'autre avion.

Un simulateur de trafic vous sera fourni, il n'y aura que l'algorithme ORCA à coder. Celui-ci est décrit plus en détail dans la section qui suit.

II. Optimal Reciprocal Collision Avoidance (ORCA)

This section describes the Optimal Reciprocal Collision Avoidance (ORCA) algorithm developed in [1] in the specific case where speed norms are constrained. First the maneuver model is detailed for an aircraft pair, then the maneuver calculation is explained when more than two aircraft are involved in a conflict.

A. Separation Constraint Model for Two Aircraft

Let us consider two aircraft A and B, as illustrated on Figure 1.

Let d be the required minimum separation and τ be a look-ahead time (or time horizon, in the rest of the paper). Consider a circle of radius d centered at aircraft B, the two lines issued from position A, tangent to the circle of radius d form a cone. If the relative speed $\overrightarrow{v_r} = \overrightarrow{v_A} - \overrightarrow{v_B}$ lies in this cone, a conflict will occur in the future. A circle of size $\dfrac{d}{\tau}$ centered at B' such that $\overrightarrow{AB'} = \dfrac{\overrightarrow{AB}}{\tau}$ defines a zone (in light red in Figure 1.b) bounded by the bold line. The red zone is denoted by δ_τ^-, and the edge of this zone is denoted δ_τ. $\overrightarrow{v_r}$ lies in this zone δ_τ^- if and only if it intersects the disc of size $\dfrac{d}{\tau}$ centered in B', i.e. if and only if there exists $\lambda \in [0:1[$ such that $\left\|\lambda\overrightarrow{v_r} - \overrightarrow{AB'}\right\| < \dfrac{d}{\tau}$ or $\left\|\lambda\tau\overrightarrow{v_r} - \overrightarrow{AB}\right\| < d$ which means that A and B are in conflict at time $\lambda\tau < \tau$. Consequently, a conflict will occur within time τ if and only if $\overrightarrow{v_r}$ lies in this light red zone δ_τ^-.

As a result, the new speeds $\overrightarrow{v_A'}$ and $\overrightarrow{v_B'}$ must always be chosen such that the resulting new relative speed $\overrightarrow{v_r'} = \overrightarrow{v_A'} - \overrightarrow{v_B'}$ outside the red zone δ_τ^-. The ORCA algorithm is based on the principle that the effort to keep the relative speed $\overrightarrow{v_r}$ out of the red area should be minimal and shared by the two aircraft. Let $\overrightarrow{c_\tau}$ be the projection of $\overrightarrow{v_r}$ on the closest edge of the red zone δ_τ. On this projected point $\overrightarrow{v_r} + \overrightarrow{c_\tau}$, the semi-plane tangent to δ_τ is noted $P^\tau_{(A,B)}$ (in light green in Figure 1.c). If $\overrightarrow{v_r'}$ is inside this semi-plane $P^\tau_{(A,B)}$, then it is outside the red zone δ_τ^-. In its original version, the necessary minimal speed modification $\overrightarrow{c_\tau}$ to keep $\overrightarrow{v_r'}$ inside $P^\tau_{(A,B)}$ is shared equally between the two aircraft and defines two semi-planes $P^\tau_{B\to A}$ and $P^\tau_{A\to B}$ (in light green in Figure 1.c) perpendicular to vector $\overrightarrow{c_\tau}$ in which aircraft A and B can choose their new speeds. If the modified speeds $\overrightarrow{v_A'}$ and $\overrightarrow{v_B'}$ are chosen in these semi-planes (Figure 1.d), the new relative speed $\overrightarrow{v_r'}$ falls inside the semi-plane $P^\tau_{(A,B)}$.

In order to define these semi-planes, let us denote $\overrightarrow{\eta_\tau}$ a unit vector normal to the curve δ_τ computed at the point $\overrightarrow{v_r} + \overrightarrow{c_\tau}$, and pointing towards the outside of δ_τ^-. The semi-plane $P^\tau_{(A,B)}$ constraining the new relative velocity vector $\overrightarrow{v_r'}$ is defined by the following equation:

$$P^\tau_{(A,B)} := \left\{ \overrightarrow{v_r'} \in \mathbb{R}^2 \mid \left(\overrightarrow{v_r'} - (\overrightarrow{v_r} + \overrightarrow{c_\tau})\right).\overrightarrow{\eta_\tau} \geq 0 \right\}$$

Likewise, the semi-planes constraint $P^\tau_{A\to B}$ and $P^\tau_{B\to A}$ for the aircraft velocities are:

$$P^\tau_{B\to A} := \left\{ \overrightarrow{v_A'} \in \mathbb{R}^2 \mid \left(\overrightarrow{v_A'} - \left(\overrightarrow{v_A} + \frac{\overrightarrow{c_\tau}}{2}\right)\right).\overrightarrow{\eta_\tau} \geq 0 \right\}$$

$$P^\tau_{A\to B} := \left\{ \overrightarrow{v_B'} \in \mathbb{R}^2 \mid \left(\overrightarrow{v_B'} - \left(\overrightarrow{v_B} - \frac{\overrightarrow{c_\tau}}{2}\right)\right).\overrightarrow{\eta_\tau} \leq 0 \right\}$$

Using the above equations defining $P^\tau_{A\to B}$ and $P^\tau_{B\to A}$, we can prove that:

$$\forall \overrightarrow{v_A'} \in P^\tau_{B\to A},\ \forall \overrightarrow{v_B'} \in P^\tau_{A\to B},\ \left(\overrightarrow{v_A'} - \overrightarrow{v_B'}\right) \in P^\tau_{(A,B)} \qquad (1)$$

As $P^\tau_{(A,B)} \cap \delta_\tau^- = \emptyset$, if $\overrightarrow{v_r'} = \overrightarrow{v_A'} - \overrightarrow{v_B'}$ is inside $P^\tau_{(A,B)}$ then it is outside the red zone δ_τ^-.

Let us quantify the insideness of a vector $\vec{v}$ in a semi-plane $P = \left\{\vec{v} \in \mathbb{R}^2 | \langle \vec{a}, \vec{v}\rangle + b \geqslant 0\right\}$ with $\|\vec{a}\| = 1$. To do that, a function $\mathrm{evalCst}$ is defined as:

$$\text{evalCst}\left(P, \vec{v}\right) = \langle \vec{a}, \vec{v} \rangle + b$$

With this definition, we have for any vector $\vec{v}$ and semi-plane P:

$$\vec{v} \in P \Leftrightarrow \text{evalCst}\left(P, \vec{v}\right) \geq 0 \tag{2}$$

There is also an insightful decomposition of $\text{evalCst}\left(P^{\tau}_{(A,B)}, \vec{v_r'}\right)$:

$$\text{evalCst}\left(P^{\tau}_{(A,B)}, \vec{v_r'}\right) = \text{evalCst}\left(P^{\tau}_{B\to A}, \vec{v_A'}\right) + \text{evalCst}\left(P^{\tau}_{A\to B}, \vec{v_B'}\right) \tag{3}$$

With this decomposition and equation 2, one can easily prove equation 1.

B. Overview of a Simulation Loop that Uses the ORCA algorithm

Algorithm 1 summarizes the ORCA algorithm applied to constant-speed aircraft (or UAVs). Time is discretized into time steps δt. As long as every aircraft has not reached its destination, every aircraft pair (i, j) is checked to calculate the semi-planes $P_{j\to i} = P^{\tau}_{j\to i}$ and $P_{i\to j} = P^{\tau}_{i\to j}$. These semi-planes are grouped in sets of constraints CstSet_i. If every aircraft i respect these sets of constraints, then there will be no conflict for the time horizon τ.

The new velocity $\vec{v_i'}$ is chosen using the function `select_speed_ORCA` $\left(\text{CstSet}_i, \vec{v_i}, \vec{v_i}^{\text{pref}}\right)$ where $\vec{v_i}^{\text{pref}}$ is the preferred velocity vector for aircraft i, directed from the aircraft current position towards its destination. This function returns the speed closest to $\vec{v_i}^{\text{pref}}$ in the admissible velocity domain, the domain that satisfies all the constraints CstSet_i. It is possible that no reachable speed are in this admissible domain. This possibility is managed inside the `select_speed_ORCA` function.

There are acceleration constraints that limits the set of reachable speed within a δt time frame. We note $\text{ReachableSpeed}\left(\vec{v}, \delta t\right)$ the set of the reachable speeds in a δt time frame from the current aircraft speed $\vec{v}$. For instance, if we consider that the aircraft speed remains constant, the only possible maneuver are turns. With this maneuver capability, $\text{ReachableSpeed}\left(\vec{v}, \delta t\right)$ will be an arc range centered on $\vec{v}$ limited by the maximum turning rate.

In Algorithm 1, the actual new speed will be the one inside $\text{ReachableSpeed}_i\left(\vec{v_i}, \delta t\right)$ that is the closest to $\vec{v_i'}$.

C. How the New Speed is Selected by ORCA

This sub-section describe the function `select_speed_ORCA` $\left(\text{CstSet}_i, \vec{v_i}, \vec{v_i}^{\text{pref}}\right)$ that selects the new speed in ORCA. This function is detailed in Algorithm 2.

The new speed $\vec{v_i'}$ is selected inside the intersection of all the semi-planes generated with the other aircraft. This intersection noted C_i in Algorithm 2. If every aircraft selects its new velocity following this rule, then no loss of separation will occur in the next τ seconds. There is an illustration of the

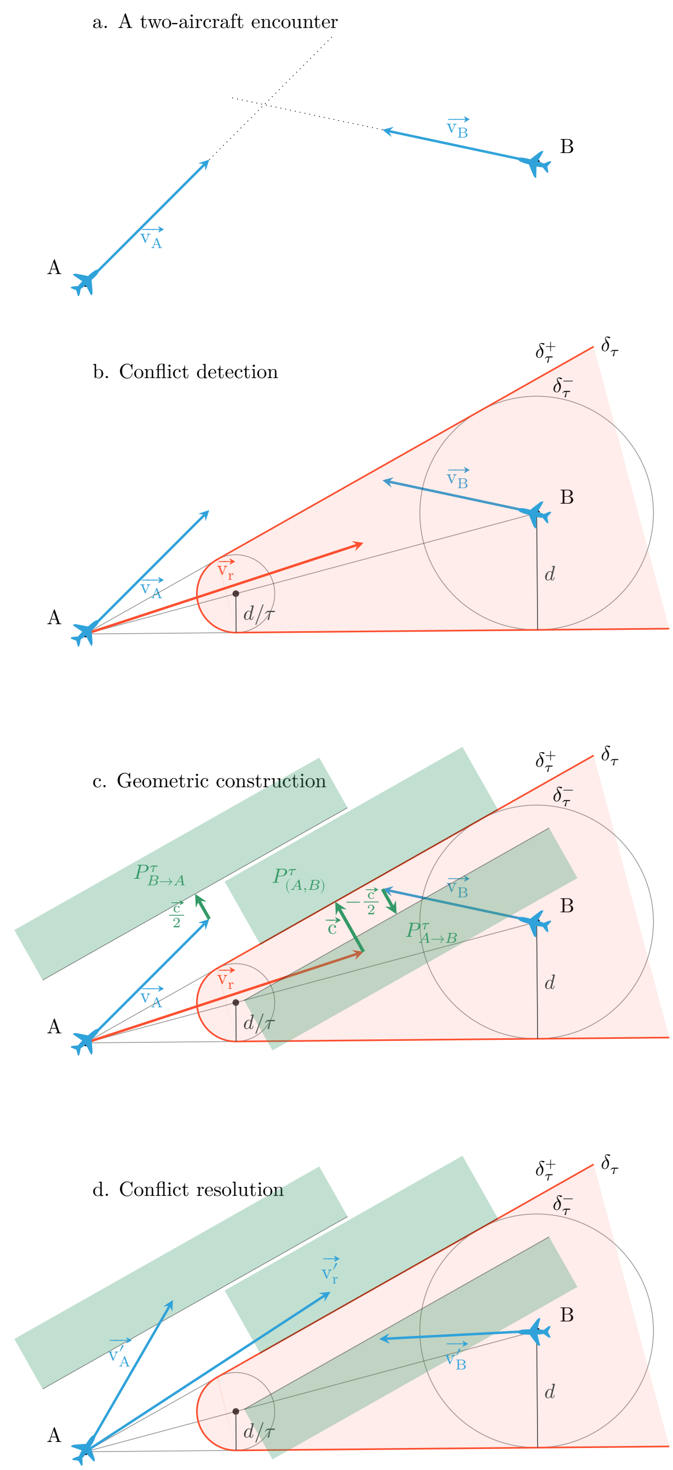

Figure 1: Conflicting aircraft model: a loss of separation will occur within time τ if and only if the relative speed $\vec{v_r}$ lies in the forbidden zone in red.

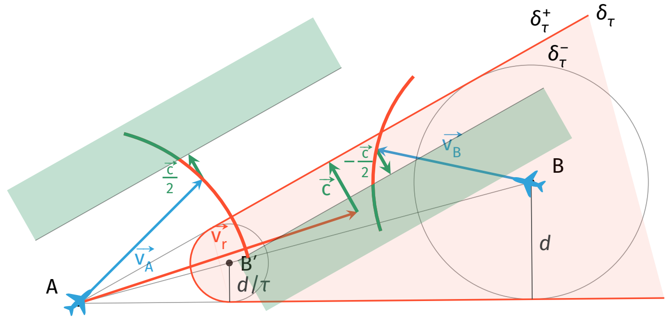

Figure 2: Conflicting aircraft model: the effort is shared by the two aircraft. The new speeds must be chosen on the green portion of the arc.

Algorithm 1 ORCA simulation loop with multiple aircraft

Input: the simulation time step δt, the anticipation τ and a scenario where aircraft start at a given positions want to reach their destinations

Output: trajectories where aircraft hopefully reach their destinations without any loss of separation

while every aircraft has not reached destination **do**
  for every aircraft couple (i,j) **do**
    Define the semi-plane constraints $P_{j\to i}$ and $P_{i\to j}$
  for every aircraft i **do**
    $\mathrm{CstSet}_i = \{P_{j\to i},\ j \neq i\}$
    $\vec{v_i'} = \mathrm{select\_speed\_ORCA}\left(\mathrm{CstSet}_i, \vec{v_i}, \vec{v_i}^{\mathrm{pref}}\right)$
    $R_i = \mathrm{ReachableSpeed}_i\left(\vec{v_i}, \delta t\right)$
    $\vec{v_i} = \mathrm{Closest}\left(R_i, \vec{v_i'}\right)$
  Move every aircraft with current speed for a δt duration
  Increase current time $t = t + \delta t$

multiple aircraft case in Figure 3 where A is conflicting with B and D. The conflict-free convex C_A for aircraft A is the intersection of the two semi-planes $P_{B\to A}$ and $P_{D\to A}$.

The velocity domain $\mathrm{ReachableSpeed}_i\left(\vec{v_i}, +\infty\right)$ contains all the velocity that can be reached by the aircraft in any amount of time. The new speed $\vec{v_i'}$ is chosen inside the intersection of C_i and the velocity domain of the aircraft $\mathrm{ReachableSpeed}_i\left(\vec{v_i}, +\infty\right)$. This intersection is noted S_i in Algorithm 2. More specifically, the selected new speed will be the one closest to $\vec{v_i}^{\mathrm{pref}}$ inside S_i.

However, S_i is empty in some situations. This happens more often when the traffic is dense. To deal with these situations, the function `empty_ORCA` described in Algorithm 3 is used. In this function, the selected new speed will be the one inside R_i^∞ that minimizes the worst violation of the constraint associated with each semi-plane in CstSet_i. This constraint violation is measured using the function `evalCst(P,` $\vec{v}$`)` which is the signed distance of the vector $\vec{v}$ to the separation plane of the semi plane P.

A geometric interpretation of `empty_ORCA` is that every semi-plane is equally slightly moved until a non-empty intersection appears. This process is described by Van den Berg [1]. Please note that in this case, as the new speed will be outside C_i, this new speed does not guarantee a conflict-free maneuver anymore, but hoppefully, it remains close to the conflict-free domain.

Algorithm 2 How the constraints are used to select the speed in ORCA

function Closest(S,$\vec{v}^{pref}$)
  return $\underset{\vec{v}\in S}{arg\,min}\ \left\|\vec{v} - \vec{v}^{pref}\right\|$
function select_speed_ORCA(CstSet_i,$\vec{v_i}$,$\vec{v_i}^{pref}$)
  $R_i^\infty = \mathrm{ReachableSpeed}_i\left(\vec{v_i}, +\infty\right)$
  $C_i = \bigcap_{P\in\mathrm{CstSet}_i} P$
  $S_i = C_i \cap R_i^\infty$
  if $S_i = \emptyset$ **then**
    return empty_ORCA $\left(R_i^\infty, \mathrm{CstSet}_i\right)$
  else
    return Closest $\left(S_i, \vec{v_i}^{pref}\right)$

Algorithm 3 How the new speed is selected when all the constraints can not be satisfied altogether.

function empty_ORCA(R,CstSet)
  return $\underset{\vec{v}\in R}{arg\,max}\ \underset{P\in CstSet}{min}\ \mathrm{evalCst}\left(P, \vec{v}\right)$

References

[1] J. van den Berg, S. J. Guy, M. Lin, and D. Manocha, "Reciprocal n-body collision avoidance," in *Robotics Research: The 14th International Symposium ISRR*, ser. Springer Tracts in Advanced Robotics (STAR), C. Pradalier, R. Siegwart, and G. Hirzinger, Eds., vol. 70. Berlin, Heidelberg: Springer, 2011, pp. 3–19.

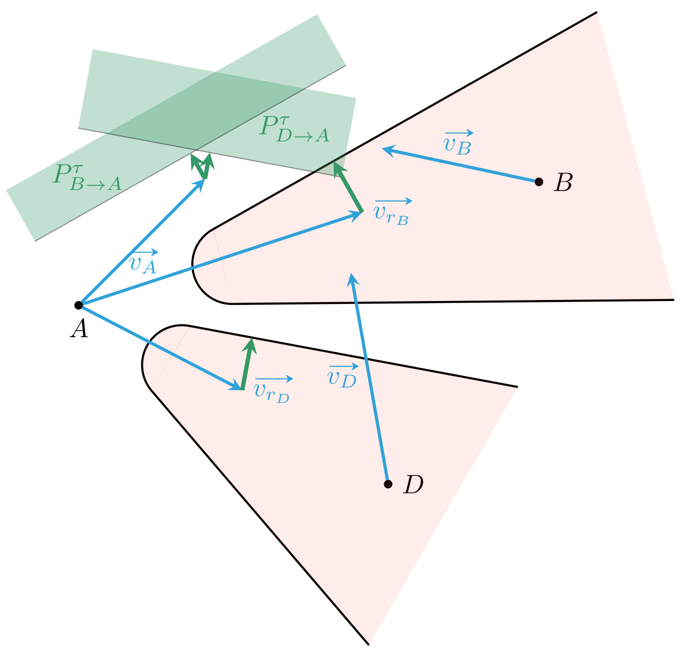

Figure 3: Multi-Conflicting aircraft model: for aircraft A, a conflict will occur within time τ with aircraft B (resp. D) if and only if the relative speed $\overrightarrow{v_{r_B}}$ (resp. $\overrightarrow{v_{r_D}}$) lies in the forbidden zone in red associated with aircraft B (resp. D).